INCH-POUND
MIL-W-23554D(SH)
31 March 1994
SUPERSEDING
MIL-W-23554C(SH)
16 March 1984
(See 6.10)

MILITARY SPECIFICATION

WASHER-EXTRACTOR, LAUNDRY, 16 TO 20 POUND CAPACITY, COMMERCIAL (SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 16 to 20 pound dry weight capacity automatic combination washer-extractor machine for use aboard ships.

1.2 Classification. The washer-extractors shall be of the following types, as specified (see 6.2):

Type I - For submarines.
Type II - For surface ships.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements For.

MIL-L-3153 - Laundry and Dry Cleaning Machinery and Equipment, Packaging of.

MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.

STANDARDS

MILITARY

MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.

MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.

DOD-STD-1399, Section 300 - Interface Standard For Shipboard Systems Section 300 Electric Power, Alternating Current. (Metric)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 123 - Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products. (DoD adopted)

A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)

A 269 - Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service. (DoD adopted)

A 366 - Standard Specification for Steel, Sheet, Carbon, Cold-Rolled, Commercial Quality. (DoD adopted)

A 525 - Standard Specification for General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process. (DoD adopted)

B 43 - Standard Specification for Seamless Red Brass Pipe, Standard Sizes. (DoD adopted)

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ASTM (Continued)

B 61 - Standard Specification for Steam or Valve Bronze Castings. (DoD adopted)

B 62 - Standard Specification for Composition Bronze or Ounce Metal Castings. (DoD adopted)

B 75 - Standard Specification for Seamless Copper Tube. (DoD adopted)

D 2000 - Standard Classification System for Rubber Products In Automotive Applications. (DoD adopted)

D 3572 - Household Laundry and Dishwasher Drain Hose, Consumer Product Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

45 - Recommended Practice for Electric Installations on Shipboard.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, Inc., 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Materials.

3.2.1 Structure. Frame, base, and strength members shall be constructed of malleable iron, ductile iron, or steel, hot-dipped galvanized in accordance with ASTM A 525 for corrosion protection or primed and painted to meet the standards specified in 3.4.5. Gray cast iron or other brittle materials shall not be used for these parts.

3.2.2 Shell. Shell shall be made of corrosion-resisting steel in accordance with ASTM A 167, type 302 or 304 or nickel-copper-aluminum alloy (minimum 60 percent nickel). Front of the shell shall be of corrosion-resisting steel metal or ferrous metal with corrosion-resisting metal liners on the side exposed to the washing solution. The shell shall be provided with a nickel-copper-aluminum alloy (minimum 60 percent nickel) or corrosion-resisting steel door frame.

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3.2.3 Cylinder. The cylinder shall be made of corrosion-resisting steel in accordance with ASTM A 167 type 302 or 304, or of nickel-copper-aluminum alloy (minimum 60 percent nickel). The driving trunnion and spider shall be of the same material as the cylinder, or of ferrous metal protected by hot-dip galvanizing in accordance with ASTM A 525, or a corrosion-resisting coating.

3.2.4 Cabinet. The cabinet shall be made of corrosion-resisting steel in accordance with ASTM A 167, type 302 or 304.

3.2.5 Metal joints. Rivets, bolts, screws, nuts, and washers shall be of steel, except when brass or corrosion-resisting metal is fastened, in which case they shall be of brass or corrosion-resisting metal. Where dissimilar metals are fastened, rivets, bolts, screws, nuts, and washers shall be galvanically compatible with materials being joined. Electrodes or welding rods used for welding shall deposit material similar to the base metal. Nuts subject to vibration or shock shall be locked. Welded steel parts, required to be galvanized, shall be galvanized after completion of welding in accordance with ASTM A 123. Drill point screws are prohibited. Spacing of fasteners shall provide strength, tightness, and rattle-free performance. Fasteners of parts through which servicing access is required shall be provided for convenient access.

3.2.6 Pipes, fittings, and valves. Parts in contact with the wash solution shall be bronze or corrosion-resisting steel. The following materials shall be used:

Pipes:	Corrosion-resisting steel, in accordance with ASTM A 269, grade TP 302 or 304.
	Brass, in accordance with ASTM B 43, grade A.
	Copper, in accordance with ASTM B 75.
Fittings:	Bronze, in accordance with ASTM B 61 or ASTM B 62.
Valves:	Cast bronze, in accordance with ASTM B 61.

3.2.7 Drain assembly. The drain assembly shall be made of the following materials:

Drain valve body:	Cast bronze, in accordance with ASTM B 61. Plastic, in accordance with the manufacturer's commercial practice.
Drain fittings:	Cast bronze, in accordance with ASTM B 61 or ASTM B 62. Plastic, in accordance with the manufacturer's commercial practice.
Drain piping:	Corrosion-resisting steel, in accordance with ASTM A 269, grade TP 302 or 304. Brass, in accordance with ASTM B 43 grade A. Copper, in accordance with ASTM B 75. Rubber hose, in accordance with ASTM D 3572.

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3.2.8 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.9 Restricted material. The tumbler-dryer shall be free from asbestos, wood, or cadmium materials.

3.3 Configuration. The type I and type II machines shall be front loading, automatic, shell and cylinder (open pocket) type with a rotating and reversing action about its horizontal axis. The machines shall have a dry weight capacity of not less than 16 pounds. Each machine shall be a complete unit, enclosed within a cabinet and consist of necessary controls, valves, motor, and appurtenances for automatically washing, rinsing, and extracting clothing as specified herein.

3.3.1 Special design requirements.

3.3.1.1 Type I washer-extractor only. The type I machine shall be so constructed that it can be readily disassembled into sections or parts that can be passed through a 25-inch diameter submarine hatch and reassembled with a minimum of effort in the submarine laundry space. Separation of parts by torch or similar means to accomplish disassembly at the site and reassembly of parts by welding, riveting, or similar method shall not be used to meet disassembly requirement. For washer-extractor installation configurations where the automatic control programmer requires remote mounting, not less than a 20 foot length of electrical cable shall be provided. The terminal ends of the connecting wiring shall be provided with connectors and receptacles for convenient installation. Connectors shall be lock type to ensure secure connection.

3.3.1.2 Type II washer-extractor only. The type II machine shall be constructed to disassemble into sections or parts that can pass through a 26 by 66 inch watertight door and be reassembled without cutting parts, welding, or using other hot work methods.

3.3.2 Detail requirements. Except as noted in the following paragraphs, all the following detail requirements apply to both type I and type II washer-extractors.

3.3.2.1 Size. The type I machine overall dimensions shall be not greater than 34 inches in height, 24 inches in width (across front) and 32 inches in depth (front to back). The type II machine overall dimensions shall be not greater than 51 inches in height, 28 inches in width (across front) and 32 inches in depth (front to back).

3.3.2.2 Weight. The machine, including controls, programmer and appurtenances, shall weigh not greater than 460 pounds.

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3.3.2.3 Water consumption. Water consumption shall be not greater than 50 gallons for a maximum rated capacity load.

3.3.2.4 Mounting. The base shall be fitted with bolt holes (not less than four) for anchoring the unit to the deck. Size of bolt holes shall be suitable for 5/8-inch bolt.

3.3.2.5 Shell. The shell shall be watertight, except in way of required openings, and rigid to prevent distortion or objectionable vibration under normal shipboard operation. Reinforcement shall be provided where necessary to properly support attachments. Provision shall be made for replacement of cylinder. Shell seams shall not pass through the drain valve opening.

3.3.2.5.1 Shell openings. The shell shall have openings for a door, drain valve, water inlet valves, cylinder shaft, water level sensor, chemical supply door, and overflow connection. No other openings shall be provided.

3.3.2.6 Cylinder. The cylinder shall be rigid to prevent distortion or failure from stresses caused during the extraction cycle at rated capacity. Perforations in the cylinder shall be sized and spaced for efficient washing, rinsing, and extraction. Perforations shall be free from burrs or imperfections that will adversely affect clothing.

3.3.2.7 Door. The shell shall be fitted with a round door mounted on heavy duty hinges, complete with neoprene gasket and heavy duty latch to ensure water-tightness when door is closed. Where door is fitted with a window, the window shall be made of tempered glass of a quality and thickness to prevent damage under normal operation. The glass shall be sealed and capable of ready replacement.

3.3.2.8 Drain valve. The drain valve shall have a resilient seat and shall be fitted at the lowest point of the shell. The valve shall be normally open.

3.3.2.9 Piping. Hot and cold water supply piping, valves and fittings shall withstand a working pressure of 100 pounds per square inch (lb/in^2).

3.3.2.9.1 Inlet valves. Fresh water inlet valves shall be connected to the machine, not less than 3 inches above the maximum high-water level.

3.3.2.9.2 Overflow connection. Machine shall be fitted with an overflow connection. The connection shall be of a size to prevent water from rising 3 inches above high-water level when both hot and cold water valves are fully opened and inlet pressure is 40 lb/in^2. The overflow connection shall be suitably located or properly diverted to prevent discharge of liquid on any electrical component.

3.3.2.10 Drive. The cylinder shall be driven by a two-speed electric motor as specified in 3.3.3.1 through V-belts to provide proper speeds for washing and extracting cycles. The cylinder shall reverse rotation not less than three times per minute during the washing cycle.

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3.3.2.10.1 Bearings. The cylinder shall be supported by antifriction bearings (see 3.3.3.1) which shall be readily replaceable and provided with seals. to prevent water entry. Bearings shall be provided in accordance with bearing contractor recommendation and shall be selected based upon the maximum bearing load during operation of the washer-extractor.

3.3.2.10.2 V-belts. V-belts shall have a service life of not less than 5000 hours of machine operation. A simple means shall be provided for properly tensioning the V-belts.

3.3.2.10.3 Shafts. Shafts and related drive items shall be constructed so that the maximum stresses are not greater than 70 percent of the yield strength of the materials under the maximum applied operating torque under full load operating conditions.

3.3.2.11 Automatic programmed control. The machine shall be equipped with a timer-programmer, preprogrammed by the manufacturer, to automatically control the fill, wash, drain, rinse, and extract operations of the machine from beginning to end of entire machine cycle. The control shall be programmed to provide not less than one wash, three rinses, and two intermediate and one final extract operations.

3.3.2.12 Water temperature control. Temperature switches shall be provided to allow operator selection of three wash water and two rinse water temperatures as follows: hot wash - warm rinse, warm wash - warm rinse, warm wash - cold rinse, cold wash - cold rinse.

3.3.2.13 Water level control. An automatic water level control shall be provided to control not less than two water levels. If the preprogrammed timer uses one water level throughout the preprogrammed wash cycle formulas, a dial or switch shall be provided for the operator to select the different preset water level settings. If the preprogrammed timer uses more than one water level throughout the preprogrammed wash cycle formulas, the preset water level settings shall be adjustable. Both automatic water level controls shall be electrically interlocked to prevent the washer-extractor program timer from advancing during the water filling operation.

3.3.2.14 Lubrication. Washer-extractors which require lubrication maintenance shall be installed with grease fittings with extension lines leading to the point of lubrication which shall be conveniently located on front or side of machine to facilitate machine lubrication from front of machine. Other points of lubrication shall be easily accessible.

3.3.2.15 Finish and color. The washer-extractor shall be finished according to contractors standard commercial practice (see 3.4.5).

3.3.2.16 Chemical supply door. Chemical supply door shall be rigidly attached to cabinet by hinges or similar means.

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3.3.3 Electrical requirements. Unless otherwise specified (see 6.2), all electrical equipment shall conform to the recommended practices for electrical installation aboard ship in accordance with IEEE 45. Electrical equipment shall meet the constraints and be constructed for operation on a type I electrical system in accordance with DOD-STD-1399, section 300. The control panel shall be dripproof. Pushbuttons and switches shall be arranged for safe and convenient operation.

3.3.3.1 Motor. Motor shall be two speed and conform to the following requirements:

Voltage:	440 volts, 3-phase, 60 hertz (Hz).
Type:	Squirrel cage induction.
Ambient temperature:	40 degrees Celsius (°C).
Duty:	Wash; reversing, intermittent, 3 reversals (minimum) per minute.
	Extract; 10 minute full load operation followed by 10 minute rest.
Bearings:	Ball.

3.3.3.2 Controls. Motor controllers shall be of the low voltage protection type. Electrical components mounted outside the control panel shall be provided with dripproof enclosures.

3.3.3.3 Wiring methods and practices. Terminal blocks shall be plainly marked to correspond with the markings on the electrical diagram. Wires run in conduit or in harness assemblies shall be marked to correspond with the electrical diagram.

3.3.4 Safety features. Exposed belts, shafts, pulleys, and other moving parts shall be fully enclosed by metal guards. Metal guards shall be firmly supported but readily removable for maintenance.

3.3.4.1 Door interlock. An electric interlock shall be provided to prevent opening door until the cylinder is at rest and prevent rotation of cylinder when shell door is opened.

3.3.4.2 Emergency stop. A means shall be provided for readily stopping the machine in an emergency.

3.4 Performance. The washer-extractor shall meet the performance requirements in 3.4.1 through 3.4.4.

3.4.1 Inclined operation. The machine shall function without reduction in performance or spill more than a total of one gallon of water when operated at an angle of 12 degrees to each side of two vertical planes at right angles to each other when operated (see 4.3.2). Spillage, if any, shall not cause damage to electrical parts.

3.4.2 Airborne noise and structureborne vibration, type I washer-extractor only. The washer-extractor shall be not greater than the acceptable airborne and structureborne vibration limits specified in tables II and III when tested in accordance with 4.3.3 through 4.3.3.7.

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3.4.3 Extraction. The machines shall extract water from saturated test material to a water retention rate of not greater than 80 percent of the normal dry weight (for instance 16 pounds of test material saturated and extracted shall weigh no more than 28.8 pounds after 5 minutes extraction). Test cloth shall contain not less than 80 percent cotton (see 4.3.4).

3.4.4 Shock, type I washer-extractor only. No part of the washer-extractor or the automatic programmer control unit shall become adrift when tested to the high impact shock tests in accordance with MIL-S-901, grade B as specified in 4.3.5.

3.4.5 Finished painted surface requirements.

3.4.5.1 Chemical resistance requirements. The test item's painted surface shall not show any changes in appearance after being exposed to the test specified in 4.3.6.1.

3.4.5.2 Hardness requirements. The test items finished painted surface shall remain undamaged when subjected to the test specified in 4.3.6.2.

3.4.5.3 Adhesion requirements. A classification of not greater than one panel out of two shall be allowed to have any of the item's finished painted surface removed when subject to the test specified in 4.3.6.3.

3.4.5.4 Humidity requirements. The test panel shall remain undamaged and shall not show any changes in appearance after being subjected to the tests specified in 4.3.6.4.

3.5 Identification and information plates. An identification plate of corrosion resistant steel or brass and an information plate of corrosion resistant steel or brass shall be secured in a conspicuous place on each machine. Marking shall be of a method which shall be permanent and legible (see 3.5.1 and 3.5.2).

3.5.1 Marking on identification plate. Marking on identification plate shall consist of the manufacturer's type of model number, serial number, contract or order number, and date of manufacture. The salient features including size, weight, speeds, voltage, current, and the national stock number.

3.5.2 Marking on information plate. Information plate shall be marked to include the following: "MACHINE NOT TO BE LOADED WITH MORE THAN RATED POUNDS (DRY WEIGHT) OF CLOTHING."

3.6 Workmanship. In general, workmanship shall be equivalent to the best in the industry and shall conform in detail to the following:

(a) Welds that come in contact with the material being processed or welds external to the machine shall be cleaned and ground to present a smooth and uniform finish. Welds internal to the machine shall be cleaned. The welds on parts and assemblies are to be cleaned or cleaned and ground, as specified above, before receiving any required protective coating to inhibit corrosion.
(b) All parts are tight and of rigid construction so that the machine does not leak or vibrate during operation.

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(c) Cylinder, with load and without, does not wobble during operation.
(d) Access plates and doors are provided as necessary for proper maintenance, cleaning and lubrication, and are of suitable size and properly located.
(e) Controls and door handle are suitably located and hazard free to operator.
(f) All surfaces are free of burrs and sharp edges.
(g) Safety guards are adequate and suitably located.
(h) Drive adjustment mechanism is suitable for the purpose intended.
(i) Bearings, gears and shafts are properly designed, alined and fitted for durable operation.
(j) Lubrication fittings are fitted where necessary and are adequate for purpose intended.
(k) Fastenings are adequate and accomplish the purpose for which intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations in table I and tests specified in 4.3.2 through 4.3.6 (see 6.3).

4.3.1 Sample machine. One machine shall be made ready for approval of the contracting officer or his authorized representative for examinations in table I and tests specified in 4.3.2 through 4.3.6 (see 6.5).

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4.3.2 Inclination test. The machine shall be inclined at an angle of 12 degrees to each side of two vertical planes at right angles to each other and shall operate with capacity load for a full cycle in each of the four positions. Water spillage shall be not greater than one gallon during the test, and the spillage shall not come in contact with electrical parts (see 3.4.1).

4.3.3 Airborne and structureborne acoustical tests; type I washer-extractor only. The washer-extractor shall be tested in accordance with MIL-STD-740-1 and MIL-STD-740-2, and as specified herein.

4.3.3.1 Test fixture and set up procedures. For the purposes of these tests the washer-extractor shall be specially mounted on a test fixture to simulate shipboard installation. Design plans for fabrication of the fixture and installation procedures shall be as specified (see 4.3.3.5).

4.3.3.2 Test conditions. For both airborne and structureborne noise tests the machine shall be loaded with 10 pounds (dry weight) of test garments and noise monitored in all modes of machine cycle. The noise level plots shall include wash, rinse, extract, and the fill modes with both inlet valves open. The entire test shall then be repeated with a 16 pound (dry weight) test load. The test garments shall contain not less than 80 percent cotton.

TABLE I. Classification of defects.

Examine	Defect (requirement paragraph)	Classification	
		Major	Minor
Materials	Construction materials not as specified (see 3.2).	X	
Configuration, general	Configuration not as specified (see 3.3, 3.3.1.1, and 3.3.1.2).	X	
Automatic programmed control	Cannot provide control as specified (see 3.3.2.11).	X	
Water temperature control	Not as specified (see 3.3.2.12).	X	
Water lever controls	Does not provide for two levels (see 3.3.2.13).	X	
Door interlock safety devices	Not provided, not as specified (see 3.3.4.1).	X	
Emergency stop button/switch	Not provided, not as specified (see 3.3.4.2).	X	

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TABLE I. Classification of defects - Continued.

Examine	Defect (requirement paragraph)	Classification	
		Major	Minor
Capacity, size, weight, water usage	Machine not designed for 16 pounds dry weight capacity (see 3.3).	X	
	Machine exceeds dimensions and weight limits (see 3.3.2.1 and 3.3.2.2).	X	
	Machine water consumption exceeds limits (see 3.3.2.3).	X	
Construction/ mechanical	Mounting holes not provided, not as specified (see 3.3.2.4).	X	
	Shell has openings for only valves and overflow vent (see 3.3.2.5.1).	X	
	Drain valve not as specified (see 3.3.2.8).	X	
	Piping, valves and fittings, not as specified (see 3.3.2.9 and 3.3.2.9.1).	X	
	Inlet valves are not 3 inches minimum above the maximum high water level (see 3.3.2.9.1).	X	
	Overflow connection not as specified (see 3.3.2.9.2).	X	
	Cylinder bearings and seals not readily replaceable (see 3.3.2.10.1).	X	
	V-belts not properly sized or not readily adjustable (see 3.3.2.10.2).	X	
	Grease line extension lines not provided (see 3.3.2.14).	X	
Construction/ electrical	Motor not as specified (see 3.3.3.1).	X	
	Electrical connection boxes are not dripproof enclosed or better (see 3.3.3.2).	X	
	Electrical terminal blocks not marked to correspond with electrical diagram (see 3.3.3.3).	X	
	Wires not properly marked to correspond with electrical diagram (see 3.3.3.3).	X	
	Length of automatic control panel cable not as specified (see 3.3.1.1).	X	
Safety features	Belts, shafts, pulleys not properly enclosed (see 3.3.4).	X	
	Door interlock safety device not provided, or as specified (see 3.3.4.1).	X	

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TABLE I. Classification of defects - Continued.

Examine	Defect (requirement paragraph)	Classification	
		Major	Minor
Identification	Identification and information plates not of specified material (see 3.5).		X
	Marking illegible (see 3.5).	X	
	Identification and information plates not provided, as specified (see 3.5.1 and 3.5.2).	X	
Workmanship	Surfaces of cylinder are not free of burrs and sharp edges (see 3.3.2.6).	X	
	Rust, toolmarks, pits, scratches, wrinkles are present (see 3.6).		X
	Welds not buffed (see 3.6).		X
	Surfaces are not free from burrs, sharp edges (see 3.6).		X

4.3.3.3 Airborne noise test. The washer-extractor mounted on the special test fixture shall be airborne noise tested to the aforementioned test conditions in accordance with MIL-STD-740-1 using the procedures for sound pressure level measurements. In addition to the octave band sound pressure levels, the overall A-weighted sound pressure level shall be obtained at each microphone position for each of the machines operating modes.

4.3.3.3.1 Sound pressure acceptance levels. For measurements taken as detailed above the octave band sound pressure levels and overall dBA level specified in table II shall be used as the basis for acceptance for airborne noise of the washer-extractor.

TABLE II. Washer-extractor airborne noise acceptance levels (maximum).

Sound pressure, dB at 1 meter RE .0002 dyne/cm^2 for octave band center frequencies, Hz									Sound pressure, dBA at 1 meter RE 0.0002 dyne/cm^2
32	63	125	250	500	1K	2K	4K	8K	
73	69	71	63	62	62	59	58	59	66

4.3.3.4 Structureborne noise test. The washer-extractor mounted on the test fixture shall be structureborne noise tested to the aforementioned test conditions in accordance with MIL-STD-740-2, except as specified herein.

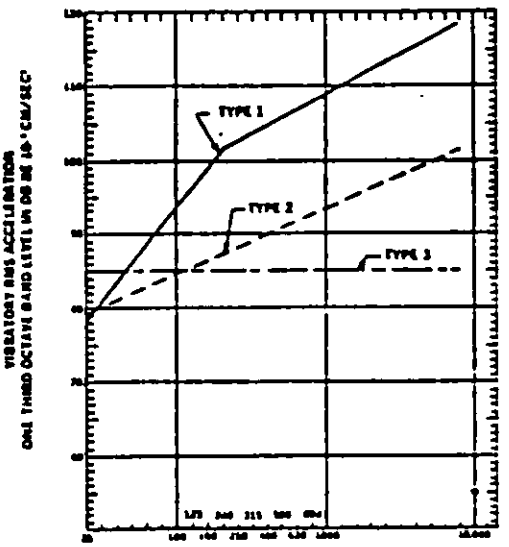
4.3.3.4.1 Selection of transducer locations. Transducer triaxial blocks shall be attached to the supporting members of the machine at not less than two diagonally opposite corners as close as feasible to the machine mounting bolts. Not less than one of the corners selected shall be the one nearest the attachment for the drive motor.

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4.3.3.4.2 One-third octave band measurements. The noise level reported for each band shall be the maximum level appearing (excluding spurious signals) in any direction for each of the machine's acoustically significant operational modes (including extraction, deep rinse, and wash-and-fill) for a wash load of 10 pounds (dry weight) of clothes, and for 16 pounds full capacity-load.

4.3.3.4.3 Structureborne noise acceptance criteria. For measurements as taken above, the type 2 one-third octave band noise level acceptance line as specified in table III shall be used for acceptance criteria.

TABLE III. Band center frequency in hertz (one-third octave).



4.3.3.5 Construction of test fixture and mounting boxes. Test fixture, equipment mounting boxes, and hold down straps shall be fabricated as specified on figures 1 and 2.

4.3.3.6 Resilient mount pads. Resilient pads shall be chloroprene polymers (neoprene) in accordance with ASTM D 2000, composition 1BC615, 1/2-inch nominal thickness, 60 durometer hardness. Pads shall be fabricated as shown on figure 3.

4.3.3.7 Acoustical test set up procedure. The test fixture shall be firmly clamped to a structural concrete floor or machinery bedplate using six strap clamps. The mounting boxes and equipment shall be mounted as shown on figure 4 and affixed to test fixture as follows:

(a) Coat 3/8-16 jacking screw with release agent, and screw into equipment mounting box jacking-screw hole so that the tip is just shy of being flush with box bottom surface. The jacking screw is for removing the mounting boxes after the test.
(b) Clean the surface of the standard test fixture table and the bottom of the mounting boxes carefully of any grease or dirt using a suitable solvent.

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(c) Clamp boxes into position on the damped table in the exact "foot-print" pattern of the machine to be tested. Check the separation distances of the box holes to be sure their spacing matches exactly the spacing of the mounting holes in the machine in all directions: transverse, fore/aft, and diagonally.
(d) Clamp blocking-fences (or pairs of locating-stops) into position along two intersecting sides of the box.
(e) Remove box hold-down clamps and coat the bottom of box with epoxy, and comb to uniform distribution.
(f) Reclamp box in the proper position against the locating stops. Remove locating stops and jacking screws before epoxy fully cures.
(g) When epoxy fully cures, remove hold down clamps. Care must be taken not to subject the boxes to shock blows, nor allow heavy components to bump them, nor to subject them to undue tensile stresses by uneven bolting procedures, by bolting to machine feet that are substantially out of plane.
(h) Machine feet should be level and in place within 0.015 inch before inserting rubber pads. Correct with annular shims.
(i) If feet are appreciably tilted or out of plane, standard 7/8-washers may be epoxied to mounting points to achieve acceptable alinement.
(j) Hold down nuts shall be tightened evenly and gradually to compress the resilient pads to a final installed height that is 73-75 percent of their initial free height.

4.3.4 Extract efficiency test. Water retention of material shall be not greater than 80 percent after 5 minutes of extraction. The test material shall contain not less than 80 percent cotton, and not less than 10 runs shall be performed to evaluate the performance of the machine under this requirement.

4.3.5 High impact shock test, type I washer-extractors only. The type I washer-extractor and it's automatic control programmer (see 3.4.4) shall be tested in accordance with MIL-S-901 and as specified herein:

(a) Shockproofness - shall be grade B.
(b) Required classes - shall be class 1.
(c) Test classifications - shall be lightweight and mediumweight for the automatic control programmer, and washer-extractor, respectively, when installation requires a remotely mounted automatic control programmer (see 3.3.1.1). Otherwise, the washer-extractor with the attached automatic controller shall be tested to mediumweight requirements.
(d) Test type - shall be type A.
(e) Failure definition - any part of test specimen items coming adrift or separating from test specimen shall constitute failure of test.
(f) Mounting - washer-extractor shall be tested on fixture shown on figure 10-1 of MIL-S-901. Automatic control programmer shall be tested on fixture 4-A as shown on figure 5 of MIL-S-901, when required to be remotely mounted.

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(g)	Mode	- equipment not operating or energized.
(h)	Disposition	- see 6.5.
(i)	Quantity for test	- 1 each washer-extractor, and 1 each automatic control programmer.

4.3.6 Test panel description. Five test panels shall be of cold rolled steel sheet, per ASTM A 366, USSG 24 gage (0.0239 inch or 0.61 mm) thick with not less than 4-1/2 inch (114 mm) width and not greater than 7-1/2 inch (191 mm) length. The five test panels shall be pretreated, prime coated, and painted in the same fashion as the equipment cabinet wall and frame members, and tested as specified in 4.3.6.1 through 4.3.6.4.

4.3.6.1 Chemical. A solution consisting of three parts deosene (Barton Solvents, Des Moines, Iowa or equivalent), one part neptune (Hopkins Agricultural Chemical Co., Madison, Wisconsin or equivalent), and one part perchlorethylene shall be used. The test panel shall be entirely submerged in the chemical solution for not less than 48 hours. After 48 hours exposure the test panel shall be wiped dry with a clean cloth (see 3.4.5.1).

4.3.6.2 Hardness. One test panel shall be subjected to a hardness test using a 2H drafting pencil. The pencil shall be sharpened to extend the lead 1/4-inch (6.4 mm) beyond the wood without reducing the diameter of the lead. The end of the lead is to be sanded against a fine sandpaper until it is flat, smooth, and circular in cross section. Holding the end of the pencil lead firmly against the finished test panel with the pencil centerline 45 degrees to the painted surface. Apply 15 gouging or scraping strokes to the finished surface. The strokes shall be not less than 1/4-inch (6.4 mm) long and applied forward to gouge or crumble the lead into the finished surface (see 3.4.5.2).

4.3.6.3 Adhesion. Adhesion between the paint and primer and between the primer and base metal of the panel shall be evaluated by a cross hatch test. Two test panels shall be scribed with a sharp knife or similar instrument to give a cross hatch pattern with an overall size of 1.0 square inch (25.4 mm by 25.4 mm block). Seventeen parallel cuts shall be made at about 1/16-inch (1.6 mm) spacing that extend through the coating to the base metal. Seventeen similar cuts shall be made 90 degrees to, and crossing the first 17 cuts; giving a cross hatched area of 256 squares. Masking tape 1 inch (25.4) wide shall be pressed firmly onto the entire hatched area and immediately jerked cleanly away from the test panel (see 3.4.5.3).

4.3.6.4 Humidity. A test panel shall be exposed to 100 percent relative humidity at 120 degrees Fahrenheit (°F) (49°C) for 168 hours. Immediately following the 168 hours of humidity testing the test panel shall undergo the hardness test in accordance with 4.3.6.2 (see 3.4.5.4).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of examination of the end item for defects specified in table I and the tests specified in 4.4.3 (see 6.3 and appendix).

4.4.1 Inspection lot. For the purpose of quality conformance inspection and test sampling, a lot is defined as all the washer-extractor machines of the same design, produced in one facility, using the same production processes and materials, and being offered for delivery at one time.

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4.4.2 Sampling for quality conformance inspection. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed units in accordance with table III and visually examine the units for the defects specified in table I. In addition, each sample selected shall be tested in accordance with the insulation resistance test provisions of 4.4.3.2. All washer-extractors in the lot shall be operationally tested in accordance with 4.4.3.1. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, the contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be examined and tested in accordance with the sampling plan contained herein. The contractor shall maintain for a period of 3 years after contract completion all records of inspections, tests, or any resulting rejections.

TABLE III. Sampling for quality conformance inspection.

Lot size	Sample size	
	Major defects	Minor defects and insulation testing
2 to 8	All	3
9 to 25	8	3
26 to 50	8	5
51 to 90	8	6
91 to 150	12	7
151 to 280	19	10
281 to 500	21	11
501 to 1200	27	15
1201 to 3200	35	18

4.4.3 Lot tests. Each sample selected in accordance with 4.4.2 shall be subjected to the following tests.

4.4.3.1 Operational testing. Each washer-extractor furnished under the specification shall be connected to electrical power supply and water, and operationally checked as follows:

(a) Automatic controller properly and automatically controls all sequences of operation in proper order for three full cycles.
(b) Door safety lock prevents opening of door during any phase of operation and engineering of cylinder when opened.
(c) No leakage is present in piping, shell, and door components.
(d) Cylinder reverses not less than three times per minute.
(e) Indicating lights, switches, pushbuttons, and timers function properly.

Failure to pass any of the above shall be cause for rejection of the unit.

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4.4.3.2 Insulation testing. Insulation resistance test shall be conducted on each machine. Insulation resistance of motor, controller, and connecting cable shall be measured by a 500-volt direct current insulation resistance tester. Insulation resistance for each component shall be not less than 10 megohms.

4.5 Inspection of packaging. Sample packages and packs and the inspection of the packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging levels. Washer-extractors shall be preserved level A or C, packed level A, B, or C, as specified (see 6.2), and marked for each level of protection in accordance with the applicable requirements of MIL-L-3153 (see 6.2). Washer-extractors shall be shipped fully assembled, unless disassembly is necessary to protect certain components during transit, or disassembly results in more than 25 percent savings in transportation cost to the Government or to the contractor. In addition for Navy acquisitions, the following applies:

(a) Navy fire-retardant requirements.

(1) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Technical manuals. Two copies of the current operation/maintenance technical manual(s) shall be packaged with each system or equipment. Technical manual packaging shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Commercial laundry washer-extractor machines acquired under this specification are intended for shipboard use. Therefore, commercial machines offered to the Government may require alteration to meet specification requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Specify whether machine is for type I or type II use (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) If first article is required (see 3.1).
(e) Electrical requirements if other than as specified (see 3.3.3).
(f) Preservation, packing, and marking requirements other than those required (see 5.1).
(g) When fire-retardant treated lumber and plywood is not required (see 5.1(a)(1)).
(h) When fire-retardant fiberboard is not required (see 5.1(a)(2)).
(i) Technical manual packaging requirements (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3 and 4.4	DI-NDTI-80603	Test procedure	----
4.4 and appendix	DI-MISC-80678	Certification/data report	10.3.2 does not apply
4.4	DI-HFAC-80272	Equipment airborne sound measurements test report	----
4.4	DI-HFAC-80274	Equipment structure-borne vibration acceleration measurements test report	----
6.7	DI-V-7002	Provisioning parts list	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4.1 Technical manuals development. Technical manuals (including technical repair standards (TRS)) should be procured under separate CLINs or subCLINs. Technical manual products (such as manuscript copy, preliminary copy, and reproducible copy) should be developed and delivered as specified in the Technical Manual Contract Requirements (TMCR) exhibit(s) to the contract. The TMCR is an exhibit or attachment to a contract specifying technical manual product development and delivery requirements, and/or management data development requirements. CDRLs and DID's should not be used for technical manual product procurements. Technical manual management data (such as plans and reports) should be procured under a separate data CLIN, but delivery should continue to be specified on CDRLs and DID's. Technical manual quality assurance management data should be prepared and delivered as specified in the TMCR and contractor order.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, and the number of items to be tested as specified in 4.3.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.7.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.8 Subject term (key word) listing.

Automatic control
Interlock
Reversing

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6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3510-N318)

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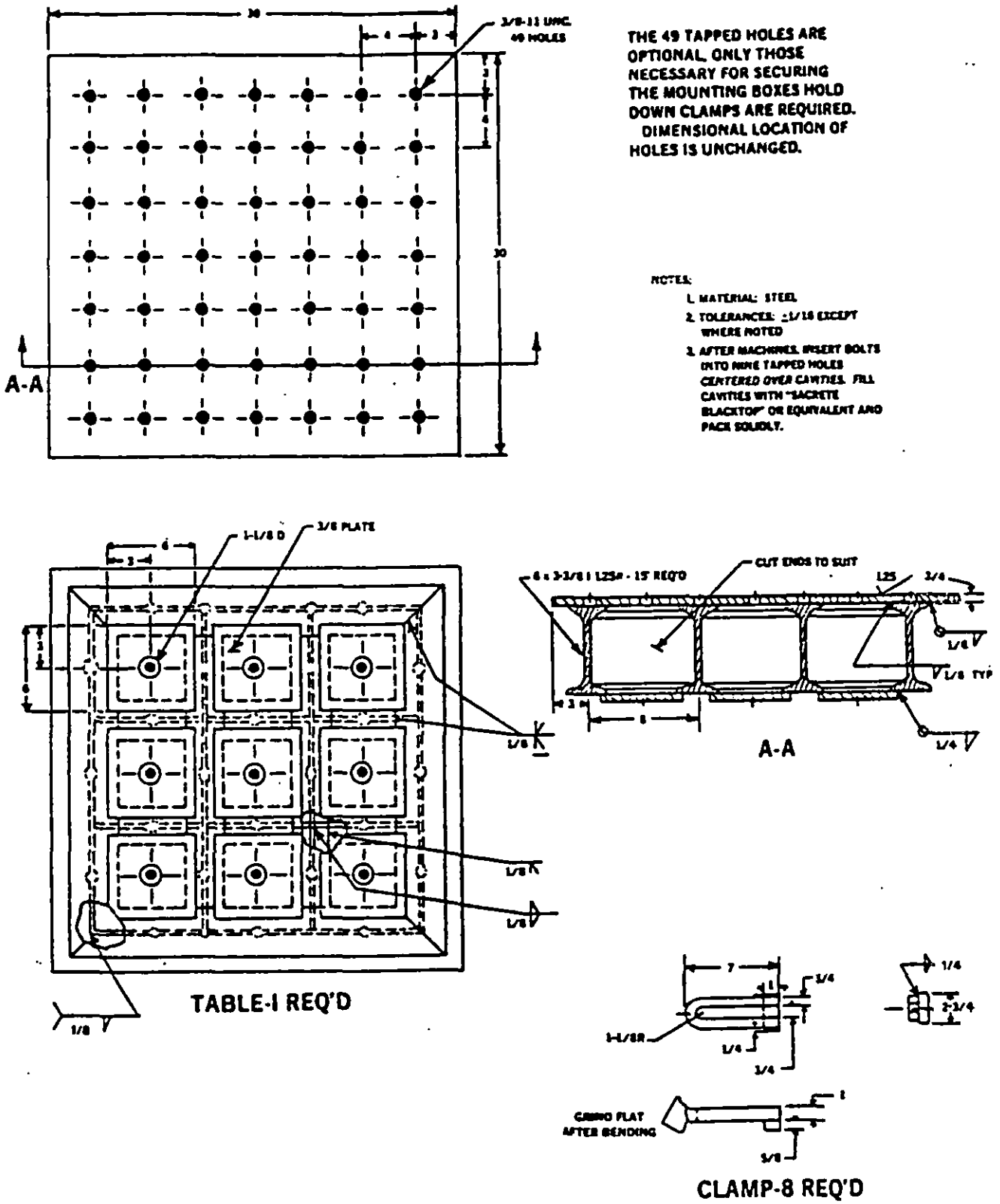


FIGURE 1. Standard test fixture and clamps.

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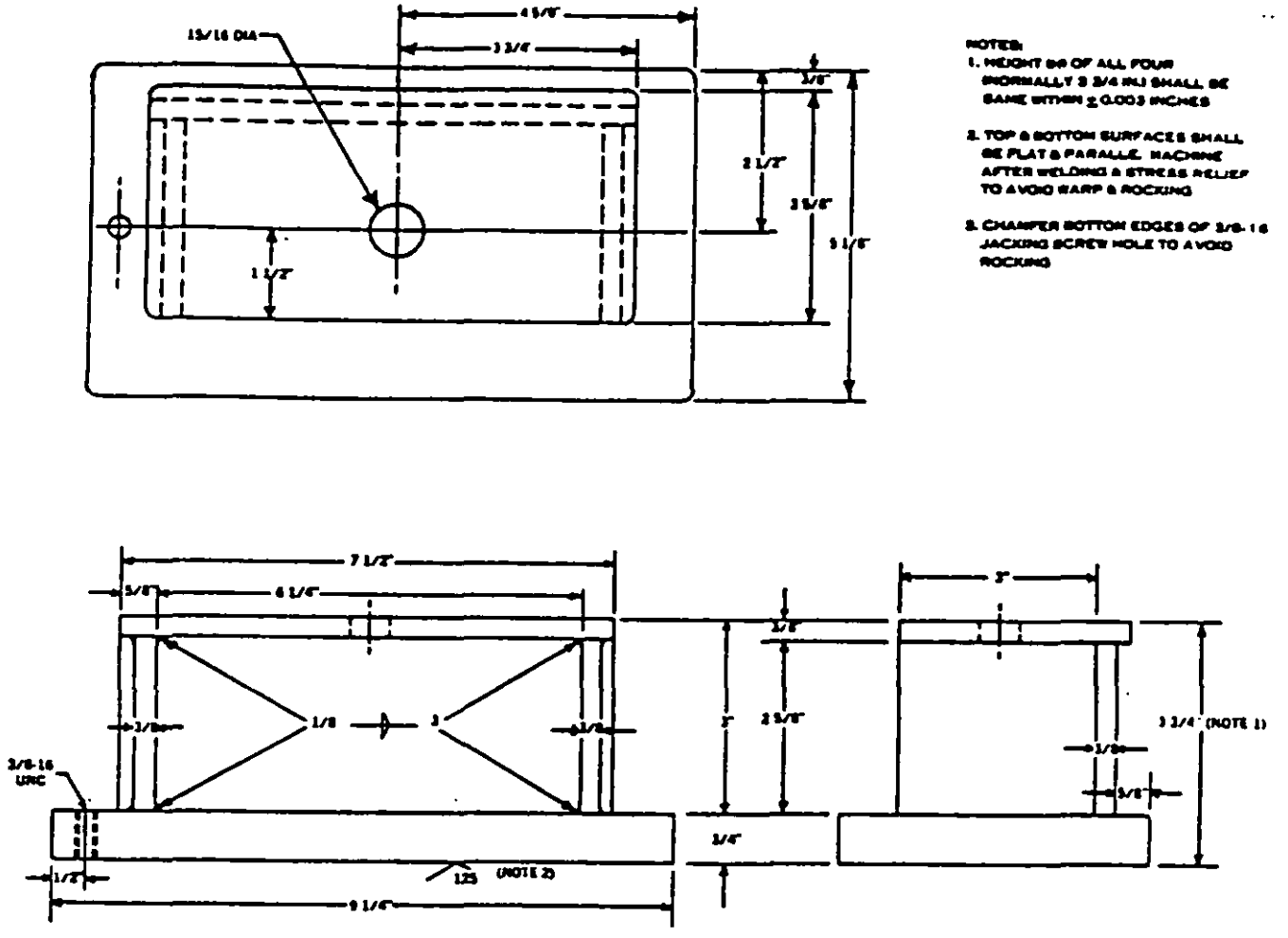


FIGURE 2. Mounting boxes - steel, 4 - required.

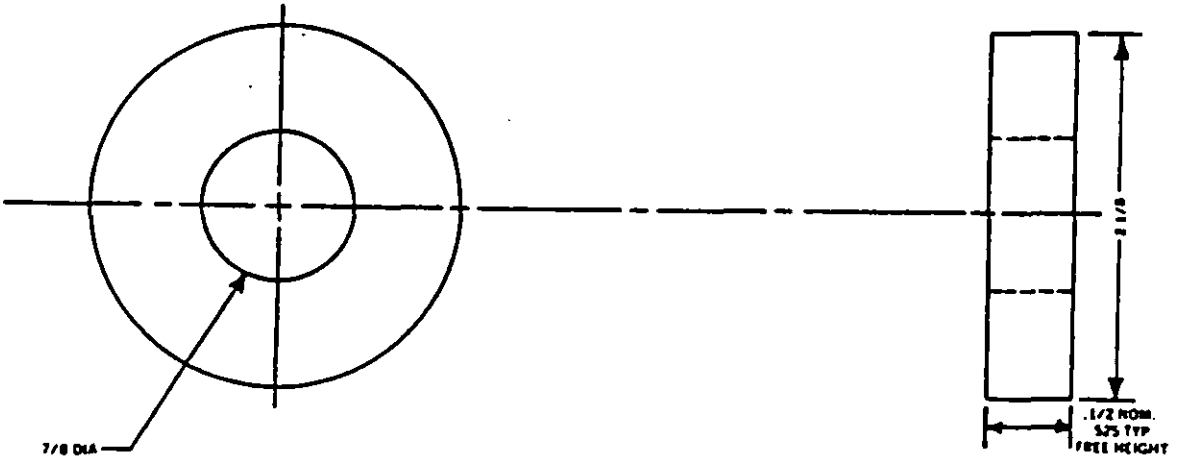


FIGURE 3. Resilient pads, 4 - required.

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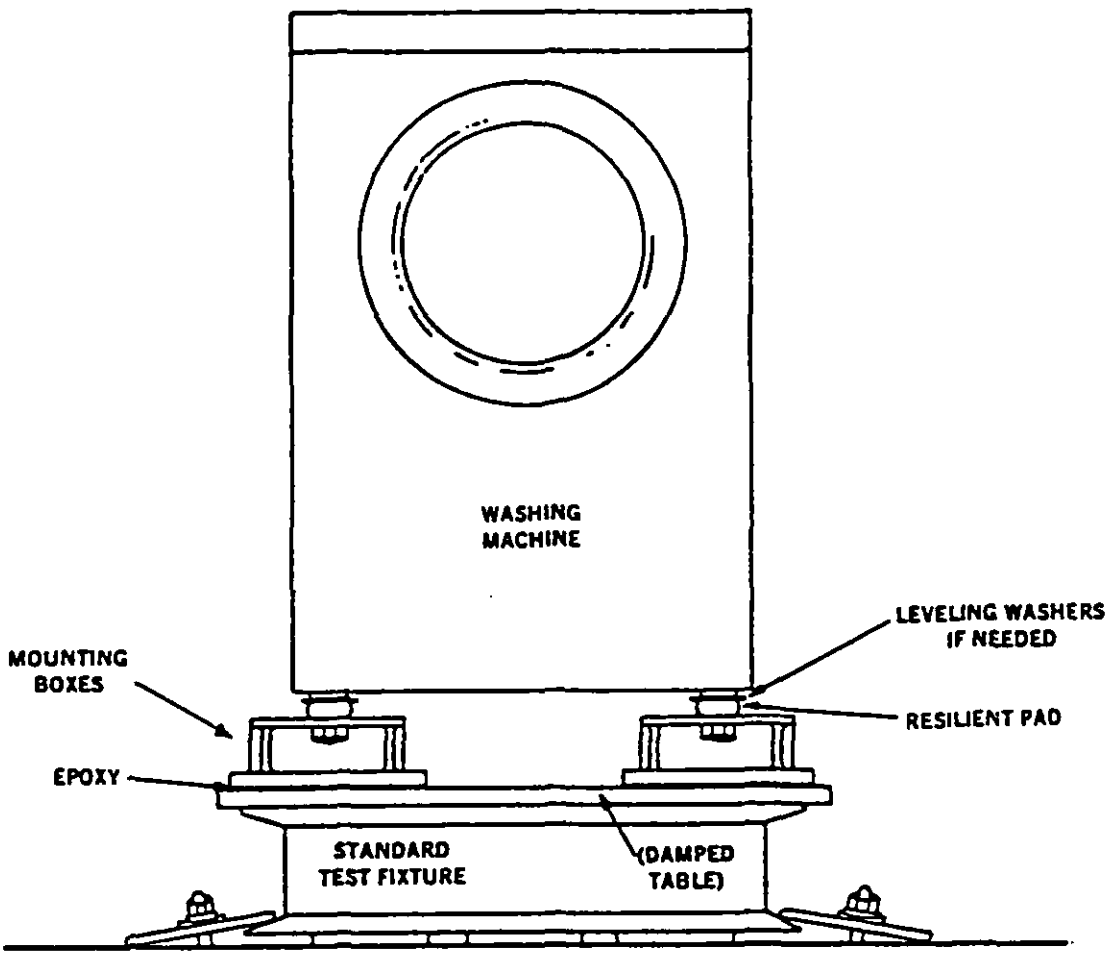


FIGURE 4. Test fixture and item assembly.

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APPENDIX

CERTIFICATION/DATA REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included in the certification/data report when specified in the contract or order. This appendix is mandatory only when data item description *DI-MISC-80678* is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. CERTIFICATION CONTENT

30.1 Test reports and certifications. The contractor shall certify in writing that the materials used in the washer-extractors are as specified herein. The contractor shall also certify that the visual, operational, and insulation resistance tests and examinations have been performed. Test conditions shall include the following conditions under which equipment was tested:

(a) Water line pressures.
(b) Composition of test garments.
(c) Type of inlet valve construction.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-23554D(SH)	2. DOCUMENT DATE (YYMMDD) 940331

3. DOCUMENT TITLE
WASHER-EXTRACTOR, LAUNDRY, 16 TO 20 POUND CAPACITY, COMMERCIAL (SHIPBOARD)

4. NATURE OF CHANGE (identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC) MR. JOSEPH J. BOWEN, NAVSSES 074F ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: TPOC: 215-897-7925 8-443-7925
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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